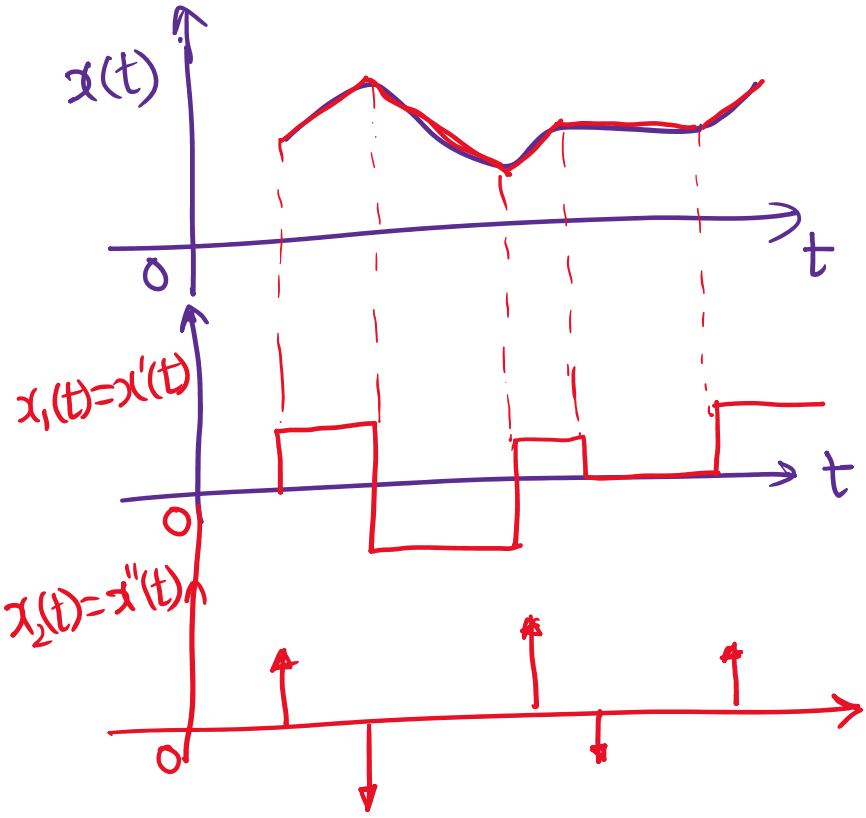
# CTFT:

$$\int_{-\infty}^{t} x(\tau)\,d\tau \longleftrightarrow \frac{X(\omega)}{j\omega} + \pi\delta(\omega)\,X(0)$$




$$\left.\begin{aligned} x_1(t) &= \int_{-\infty}^{t} x_2(\tau)\,d\tau \\ \therefore X_1(\omega) &= \frac{X_2(\omega)}{j\omega} + \pi\delta(\omega)X_2(0) \end{aligned}\right\}$$

$X_2(\omega)$ can be easily found (F.T. of impulses)

For finite durat$^n$ $x(t)$:

$$X_2(0) = \int_{-\infty}^{\infty} x_2(t)\,dt \quad \& \quad x_1(t) = \int_{-\infty}^{t} x_2(\tau)\,d\tau \Rightarrow x_1(\infty) = \int_{-\infty}^{\infty} x_2(\tau)\,d\tau$$

$\therefore X_2(0) = x_1(\infty)$ & $x_1(\infty) = 0$ ($x(t)$ is finite durat$^n$)

$\therefore X_2(0) = 0$

$$\therefore X_1(\omega) = \frac{X_2(\omega)}{j\omega}$$

Similarly, $X(\omega) = \dfrac{X_1(\omega)}{j\omega} + \pi\delta(\omega)\,\overset{x(\infty)=0}{X_1(0)} = \dfrac{X_1(\omega)}{j\omega} = \dfrac{-X_2(\omega)}{\omega^2}$

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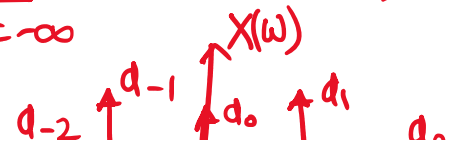
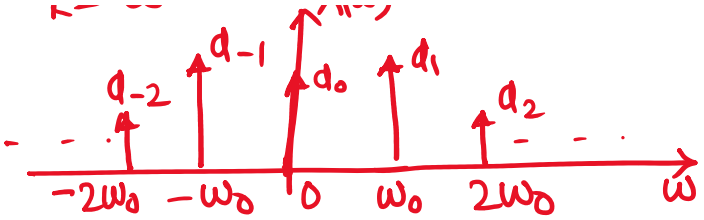
**Ex.2** Let $x(t)$ be periodic, $X(\omega) = ?$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longleftrightarrow \sum_{k=-\infty}^{\infty} a_k\, 2\pi\,\delta(\omega - k\omega_0) = X(\omega)$$

$$1 \cdot e^{j\omega_0 t} \longleftrightarrow 2\pi\,\delta(\omega - \omega_0)$$



$$1 \cdot e^{j\omega_0 t} \longleftrightarrow 2\pi\,\delta(\omega-\omega_0)$$

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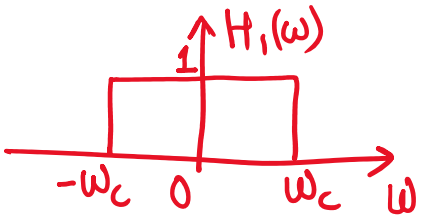
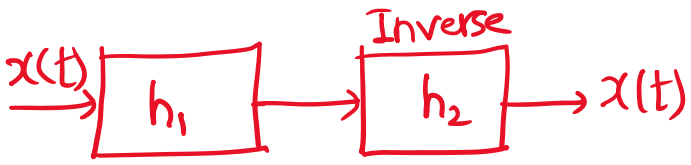
**Ex. 3**




Find inverse of $h_1$ if exists.




$\Rightarrow h_1(t) * h_2(t) = \delta(t)$

$\Rightarrow H_1(\omega)\,H_2(\omega) = 1 \Longrightarrow H_2(\omega) = \underbrace{\frac{1}{H_1(\omega)}}_{\text{Undefined for } \omega > \omega_c}$

$\therefore$ Non-invertible

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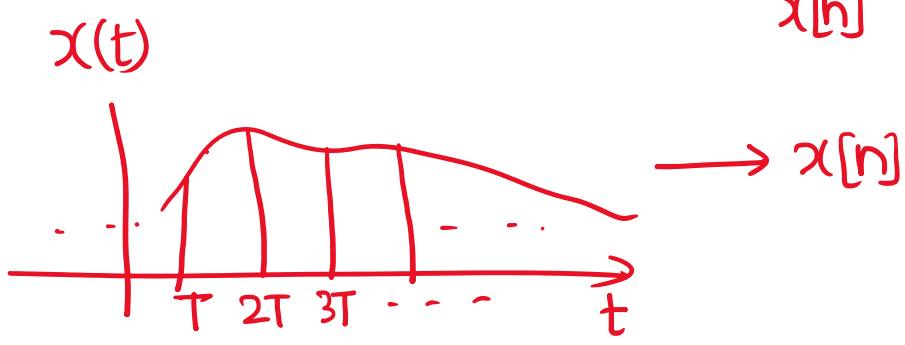
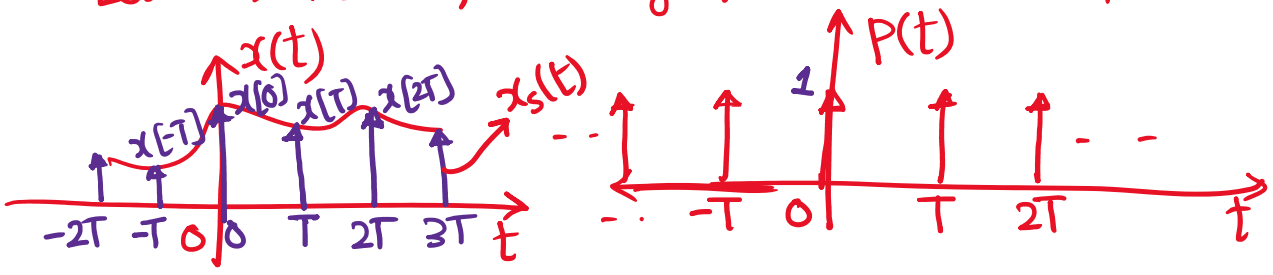
## Sampling:




## I) Ideal Impulse Sampling:

Let $x(t)$ be sampled using $p(t)$: train of impulses.




$x_s(t) \triangleq x(t)\,p(t)$

$X_s(\omega) = \frac{1}{2\pi} X(\omega) * P(\omega)$ , $P(\omega) = ?$

$x_s(t) = x(t)\,p(t)$

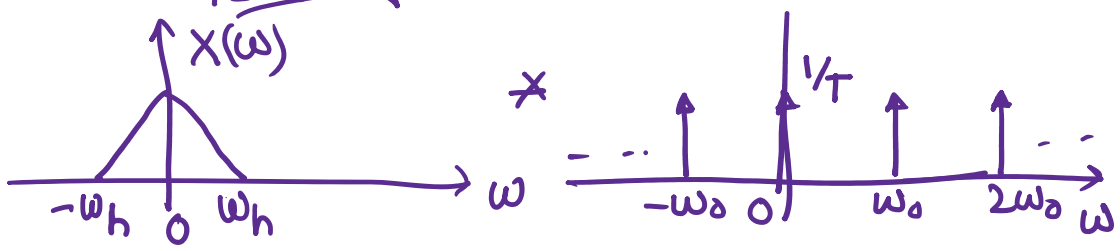
$$X_s(\omega) = \frac{1}{2\pi} X(\omega) * P(\omega) \quad , \quad P(\omega) = ?$$

$$p(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad , \quad a_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t}\,dt = \frac{1}{T} \quad \forall k$$

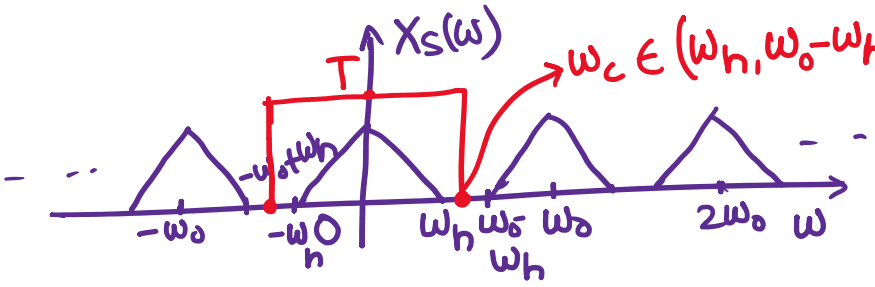
$$\Rightarrow p(t) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t} \longleftrightarrow P(\omega) = \frac{1}{T} 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

$$\therefore X_s(\omega) = \frac{1}{\cancel{2\pi}} \cdot \frac{\cancel{2\pi}}{T} X(\omega) * \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

Let $X(\omega)$ :




$\Rightarrow$




$\omega_c \in (\omega_h, \omega_0 - \omega_h)$ , $\omega_0$ : sampling freq.

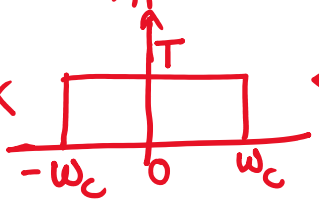
$\Rightarrow$ If $\omega_0 - \omega_h > \omega_h$ } No overlap

i.e. $\omega_0 > 2\omega_h$

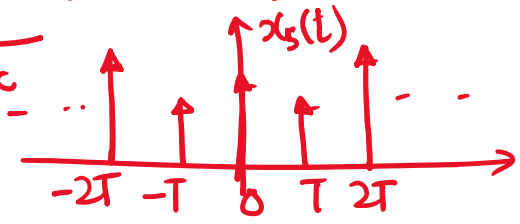
## Nyquist Sampling Theorem :

For a signal bandlimited to $f_H$, the original signal can be perfectly reconstructed from its sampled version, provided sampling freq. $f_s > 2f_H$.

Reconst$^n$ : $X_s(\omega) \times$




$\overset{\simeq}{\longleftrightarrow}$ $x_s(t) * \text{sinc}(\ )$




$* \text{sinc}(\ )$

$\therefore$ sinc( ) acts as an ideal

∴ sinc( ) acts as an ideal interpolator.

## Non-Realizable:

1) Ideal impulse can't be realized.
2) " LPF " " .
3) $x(t)$ can't be bandlimited (as it is timelimited in practice)